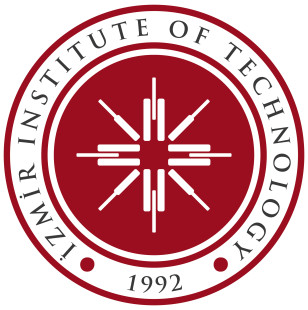


İzmir Institute of Technology

Faculty of Engineering

Department of Electrical and Electronics Engineering

EE 315 Electronics Laboratory Prelab Sheet

Experiment 2

Submitted By:

Baran Güvendiren - 310206029
Ozan İnan - 330206062
Mohammadhossein Nafaryeh - 320206059

October 17, 2026

Preliminary Work

Problem 1. Full-Wave Bridge Rectifier

a) Positive half cycle: D1 and D3
Negative half cycle: D2 and D4

b) 100 Hz because the Full wave rectifier folds the negative voltage to positive so the VL frequency is twice the input.

c) Peak voltage is 9 volts (Vs=9) so according to $2V_{sp}/\pi$ formula the average voltage is 5.730v but because of the diodes which are in series the total voltage drops by 1.2 volts so the average voltage drops to 4.530 volts. The the DC load current can be calculated with V_{LDC}/R_L which is equal to 3.775mA. In the end the total power will be calculated as $2 \cdot V_D \cdot I_{LDC} = 2 \cdot 0.6\text{ V} \cdot 3.775\text{mA} = 4.53\text{mW}$.

Problem 2. Full-wave Rectifier with a Center-tapped Transformer

1. Average DC Load Voltage: $V_{LDC} = 18/\pi - 0.6 = 5.130\text{V}$
2. Average DC Load Current: $I_{LDC} = 5.130/1200 = 4.275\text{ mA}$
3. Total Power: $1 \cdot V_D \cdot I_{LDC} = 0.6 \cdot 4.275 = 2.57\text{ mW}$

Problem 3. Full-wave Bridge Rectifier with Triangular Approximation

a) Load Regulation ($R_L = 330$ ohms)

1. **No-load condition (V_{NL}):** When R is infinity no current is drawn so the capacitor does not charge and voltage remains at peak.
   $V_{NL} = V_{LP} = 7.8\text{V}$
2. **Full Load Condition:** $f_r R_L C = 100 \cdot 330 \cdot 22 \cdot 10^{-6} = 0.726$
   $V_{FL} = 7.8/(1 + (1/(2 \cdot 0.726))) = 4.619\text{V}$
3. **Calculate Load Regulation:**
4. Load Reg $= \left(\frac{V_{NL} - V_{FL}}{V_{NL}}\right) 100\% = \left(\frac{7.8 - 4.619}{7.8}\right) \cdot 100\% = 40.78\%$

b) Calculate Ripple Factor

- **For Load Resistance = 330 ohms:**
  Ripple Factor = $1/(3.4641 \cdot 100 \cdot 330 \cdot 0.000022)$
  Ripple Factor = 0.3976
  Result = 39.76%
- **For Load Resistance = 1.2 kohms (1200 ohms):**
  Ripple Factor = $1/(3.4641 \cdot 100 \cdot 1200 \cdot 0.000022)$
  Ripple Factor = 0.1093
  Result = 10.93%
- **For Load Resistance = 10 kohms (10000 ohms):**
  Ripple Factor = $1/(3.4641 \cdot 100 \cdot 10000 \cdot 0.000022)$
  Ripple Factor = 0.0131
  Result = 1.31%

c) Effect of Load Resistance on Ripple Voltage:
As the load resistance increases, the load draws less current from the circuit. A lower current draw means the filter capacitor discharges much slower between the active charging cycles. Because the capacitor loses less

stored voltage before the next pulse arrives, the peak-to-peak voltage drop (the ripple) is significantly reduced, resulting in a cleaner, smoother DC output.

Problem 3. Full-Wave Bridge Rectifier without Triangular Approximation

Because the time constant $\tau = R_L C_1 = 330\,\Omega \times 22\,\mu\text{F} = 7.26$ ms is shorter than the 10 ms ripple period, the capacitor discharges deeply. The standard triangular ripple approximation ($V_{rplPP} = \frac{V_{LDC}}{fR_LC}$) fails under these large-ripple conditions. We must use the exact exponential discharge method.

a) Load Regulation
The peak load voltage is $V_{LP} = 9\text{ V} - 2(0.6\text{ V}) = 7.8$ V. At no-load ($R_L \to \infty$), the capacitor does not discharge, so $V_{NL} = V_{LP} = \mathbf{7.8}$ V.

At full load ($R_L = 330\,\Omega$), the capacitor voltage decays exponentially from the peak at $t = 0$:

$$v_C(t) = V_{LP} e^{-t/\tau} = 7.8 e^{-t/7.26\text{ ms}}$$

The rectifier diodes turn back on when this decaying voltage intersects the rising edge of the next rectified half-sine wave, given by $v_{\text{rect}}(t) = -9\cos(100\pi t) - 1.2$. Setting them equal:

$$7.8 e^{-t/7.26\text{ ms}} = -9\cos(100\pi t) - 1.2$$

Solving this transcendental equation numerically yields an intersection time of $t \approx 6.62$ ms. Substituting this back into the discharge equation gives the minimum trough voltage:

$$V_{\text{min}} = 7.8 e^{-6.62/7.26} \approx 3.13\text{ V}$$

The average DC voltage is approximately the midpoint of the peak and trough:

$$V_{FL} = V_{LDC} \approx \frac{V_{LP} + V_{\text{min}}}{2} = \frac{7.8 + 3.13}{2} = 5.465\text{ V}$$

$$\%\text{ Load Regulation} = \frac{V_{NL} - V_{FL}}{V_{NL}} \times 100\% = \frac{7.8 - 5.465}{7.8} \times 100\% = \mathbf{29.93}\%$$

b) Ripple Factor
The peak-to-peak ripple voltage is $V_{rplPP} = 7.8\text{ V} - 3.13\text{ V} = 4.67$ V. Using the definition from the manual:

$$\text{Ripple Factor} = \frac{V_{rplPP}}{2\sqrt{3}V_{LDC}} = \frac{4.67}{2\sqrt{3}(5.465)} = \frac{4.67}{18.93} = \mathbf{24.67}\%$$

(Note: As R_L increases to 1.2 kΩ and 10 kΩ, the time constant τ becomes very large, and the standard linear approximation becomes accurate again, yielding $\approx 10.9\%$ and $\approx 1.3\%$ respectively.)

Problem 4. Capacitive Filtering

a) Load Regulation with Added Capacitor

Adding a second $22\,\mu$F capacitor in parallel makes $C_{tot} = 44\,\mu$F. The no-load voltage remains $V_{NL} = 7.8$ V. The new time constant is $\tau = 330\,\Omega \times 44\,\mu\text{F} = 14.52$ ms. Setting the new exponential discharge equal to the rectified wave:

$$7.8e^{-t/14.52\text{ ms}} = -9\cos(100\pi t) - 1.2$$

Solving numerically gives $t \approx 7.28$ ms. The new trough voltage is:

$$V_{\text{min}} = 7.8e^{-7.28/14.52} \approx 4.72\text{ V}$$

$$V_{FL} = V_{LDC} = \frac{7.8 + 4.72}{2} = 6.26\text{ V}$$

$$\%\text{ Load Regulation} = \frac{7.8 - 6.26}{7.8} \times 100\% = \mathbf{19.74}\%$$

b) Effect of Increasing Filter Capacitance

Increasing the filter capacitance directly increases the *RC* time constant. This slows the voltage decay rate during the load discharge phase, preventing the voltage from dipping as low before the next cycle. The result is a significantly reduced ripple voltage (4.67 V down to 3.08 V) and an improved load regulation percentage.

5. LTspice Simulation

5.1 Full-Wave Bridge Rectifier

The purpose of a full-wave bridge rectifier circuit is to convert an AC voltage to a DC voltage. As the bridge rectifier rectifies both half-cycles of the input signal into a single polarity, the frequency of the output voltage is doubled relative to the input frequency. Since two diodes conduct in series during each half-cycle, the circuit has a voltage drop equal to the sum of the forward voltage drops of the two conducting diodes. To prevent severe voltage sag, a filter capacitance is needed. An example circuit is given in Figure 1.

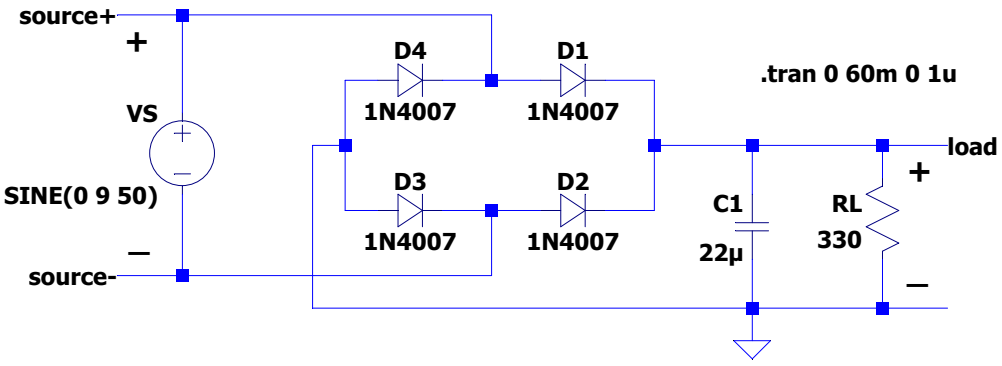


Figure 1: Schematic of full-wave bridge rectifier circuit

The input and load voltage of the example circuit is given in Figure 2.

R_L	$V_{L,\text{peak}}$	f_L	**Number of conducting diodes**
$330\,\Omega$	$7.59\,\text{V}$	$100\,\text{Hz}$	2

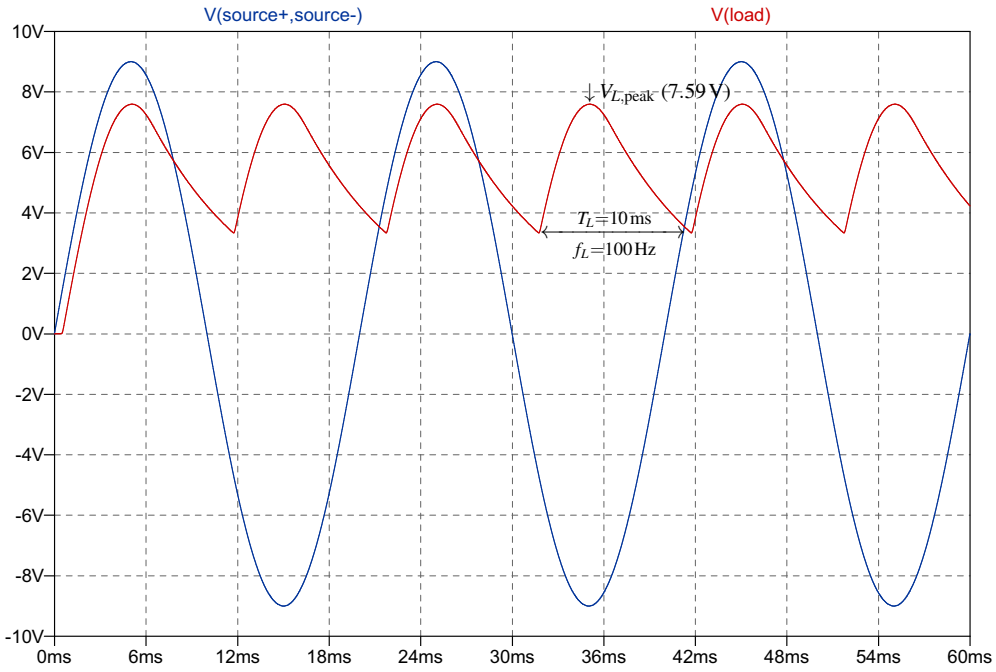


Figure 2: Input and output voltage waveforms of the full-wave bridge rectifier circuit ($R_L = 330\,\Omega$)

5.2 Zener Diode Regulator

In full-wave bridge rectifier circuits, a key limitation is that the peak output voltage ($V_{L,\text{peak}}$) and the peak-to-peak ripple voltage ($V_{rpl,PP}$) are affected by load conditions. As the load current increases, the voltage drops across the bridge diodes and series components increase, causing the output voltage to sag. To address this limitation and clamp the peak output voltage to a desired, more stable level, a Zener diode regulator can be implemented. However, since the excess voltage is entirely dropped across the series resistor (R_S), this circuit suffers from poor thermal and power efficiency, especially in high-current applications. Despite this inefficiency, it remains a low-cost practical solution for low-current voltage references and basic low-power circuits. An example circuit is given in Figure 3.

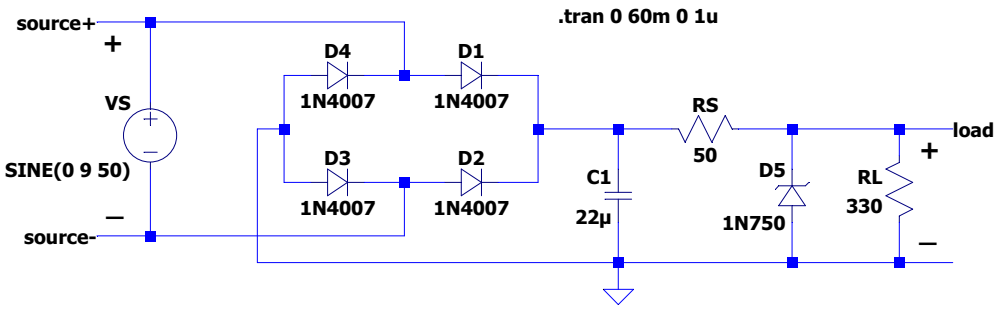


Figure 3: Schematic of Zener diode regulator circuit

The input and load voltage of the example circuit is given in Figure 4.

Breakdown Voltage (V_Z)	$V_{o,DC}$	I_Z
4.7 V	4.13 V	−13.15 mA

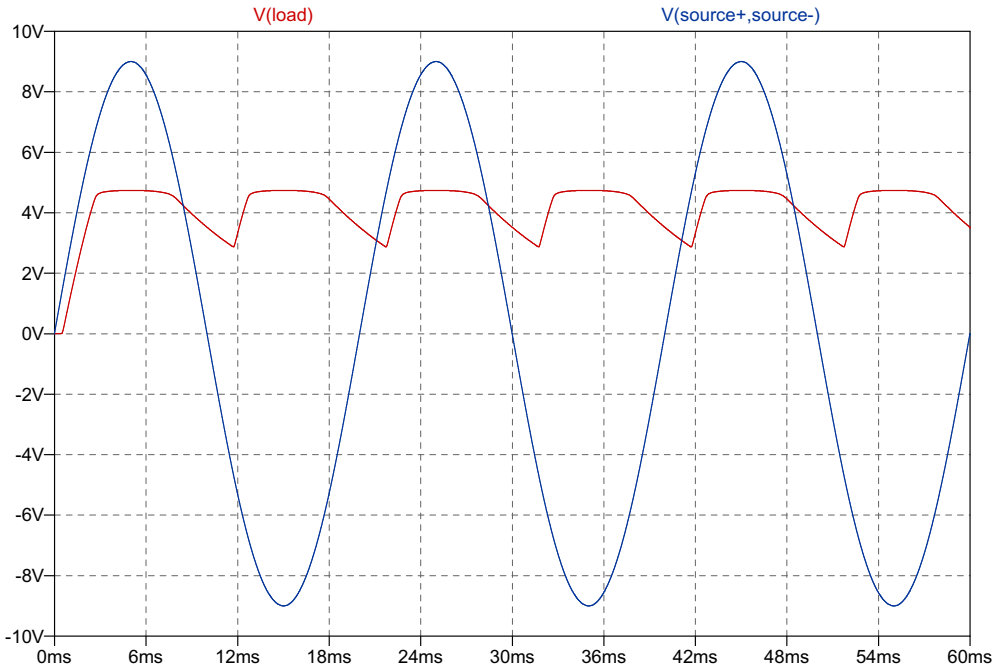


Figure 4: Input and output voltage waveforms of the Zener diode regulator circuit ($R_L = 330\,\Omega$)